Branch Specialization Checkpoint

Toy Competition Layout Permutations

Branch Specialization Project

2026-05-22

Checkpoint reason: new result refines the Phase 3 architectural-intervention claim.

Current Question

Phase 3 RQ: Does structural heterogeneity cause more stable functional specialization?

After the local-vs-induction competition result, the sharper question is:

> *When the 64-dim head is moved to each possible head position, do functions follow dimension, index, or broader layout?*

This tests whether heterogeneous head dimensions create semantic role classes, or just break permutation symmetry by making stable capacity slots.

Experiment Setup

- Task combines local copy and induction-style continuation in one sequence.
- Local region: `[x, SEP, x]`, scored at `SEP`.
- Induction region: `[y1, ..., y16, y1, ..., y16]`, scored on second-half continuation.
- Same total attention dimension: 128.
- Five seeds per layout, 1200 training steps per seed.
- Role score is causal single-head ablation loss delta.

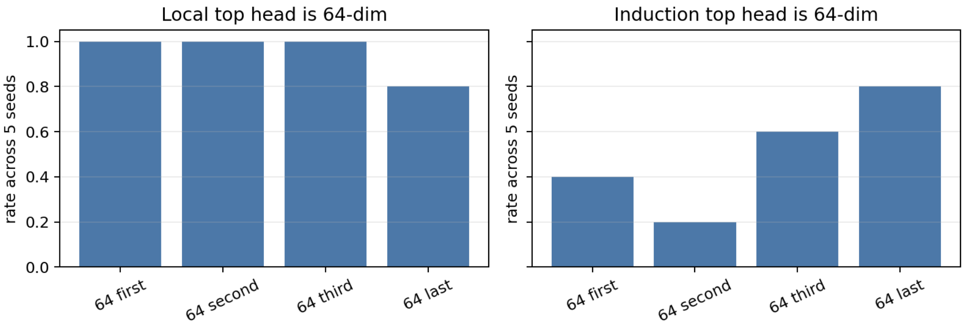
$$S_r(h) = \max(\mathrm{loss}_r(\mathrm{ablate}\ h) - \mathrm{loss}_r(\mathrm{base}), 0)$$

Layouts Tested

Config	Head dimensions	Local acc.	Induction acc.
`hetero4_64first`	[64, 16, 16, 32]	0.9999	0.9986
`hetero4_64second`	[16, 64, 16, 32]	1.0000	1.0000
`hetero4_64third`	[16, 32, 64, 16]	1.0000	0.9993
`hetero4`	[16, 16, 32, 64]	1.0000	0.9992

All layouts learn both objectives, so the slot pattern is not a failure-to-learn artifact.

Main Evidence



- Local role almost always occupies the 64-dim head.
- Induction role is layout-sensitive and only uses the 64-dim head in half of runs.

Aggregate Result

Role	Models	Top head 64-dim	Mean top spec.	Mean top loss delta
Local	20	19/20	0.9180	0.1457
Induction	20	10/20	0.7388	2.5272

The local role follows the 64-dim head across positions H0, H1, H2, and H3. Induction splits evenly between 64-dim and 16-dim top slots.

Interpretation

Supported:

- Heterogeneous head dimensions create stable high-capacity slots.
- The local/previous-token role strongly prefers the 64-dim slot.
- The function-to-slot mapping is much more stable than uniform 4-head baselines.

Not supported:

- A simple semantic taxonomy where small heads become local and large heads become induction/global.
- Dimension alone determining role assignment.

Decision

Narrow the Stage 3 claim:

> *Structural heterogeneity can stabilize function-to-slot mappings by breaking permutation symmetry, but task pressure and layout mediate which function gets each slot.*

Next decisive experiment:

1. Sweep local-vs-induction objective weights on the same task.
2. Test whether reducing local pressure lets induction occupy the 64-dim slot more often.
3. If roles flip with weights, frame the mechanism as capacity-slot competition.

Sources and Provenance

- Report: `doc/phase3_toy_competition_layout_permutations.md`
- Training script: `scripts/toy_competition_head_dim_intervention.py`
- Analysis script: `scripts/analyze_competition_layout_permutations.py`
- New run: `results/phase3_toy_competition_layout_permutations/`
- Combined analysis: `results/phase3_toy_competition_layout_analysis/`
- Deck data snapshot: `presentations/2026-05-22-0858-layout-permutation/data/`